Research Article **Open Access**

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Evaluation of change in heart rate variability due to different soundscapes

https://doi.org/10.1515/noise-2022-0158
Received Dec 02, 2022; accepted Dec 20, 2022

Abstract: Soundscapes affect the health and quality of life of humans. Noisy soundscapes have a negative impact on humans causing annoyance, sleep disturbance and cardiovascular issues. This paper analyses the change in heart rate variability (HRV) due to exposure of different soundscape stimuli. A total of 40 soundscape stimuli were collected from New Delhi India, which were grouped into three clusters, 'Loud', 'Active' and 'Silent', based on psychoacoustic indicators. Listening experiments were conducted with 25 healthy participants, during which electrocardiography responses were collected as response variable.

HRV analysis was performed to analyse the change in time domain (Heart rate, SDNN, NN50, pNN50) and frequency domain (VLF, LF, HF, LF/LF ratio) parameters. A significant change in heart rate is observed with an increase in loudness of stimuli. The change in HRV is analysed by considering noise sensitivity level of participants. A significant decrease in SDNN is noted for participants with high noise sensitivity. Frequency domain parameters of HRV did not exhibit a significant change due to noise exposure. A significant decrease in SDNN suggests imbalanced autonomic nervous system activation, which increases the risk of cardiovascular diseases, particularly for people with high noise sensitivity.

Keywords: Soundscapes; hear rate variability; noise exposure; physiological signals; electrocardiography (ECG)

Abbreviations:

ANS: Autonomic nervous system
ECG: Electrocardiogram
HF: High frequency (refers to HRV frequency band, by default 0.15-0.4 Hz)
HR: Heart rate
HRV: Heart rate variability
IBI: Inter-beat-interval (same as RR interval)
LF: Low frequency (refers to HRV frequency band, by default 0.04-0.15 Hz)
NN50: Number of successive RR interval pairs that differ more than 50 ms
pNN50: Relative number of successive RR interval pairs that differ more than 50 msec
PNS: Parasympathetic nervous system
RMSSD: Root mean square of successive RR interval differences
RR: Time interval between successive ECG R-waves (RR interval, same as IBI)
SDNN: Standard deviation of normal-to-normal RR intervals
SNS: Sympathetic nervous system
VLF: Very low frequency (refers to HRV frequency band, by default 0-0.04 Hz)
SL: Sensitivity Level (refers to high and low noise sensitivity of the listener)

1 Introduction

Sound plays an important role in defining the character of urban spaces. Soundscape is the acoustic environment perceived or experienced by people in a given context [1]. It contains different types of noise sources that occur simultaneously or at varying periods. These sounds may be categorised/perceived as positive (pleasant, e.g., birds, water, natural sound) or negative (unpleasant e.g., Traffic, honking, etc) [2]. While the positive influence of soundscapes on public health has been addressed by Aletta et al. [3], the adverse effects of loud soundscapes on human is a worldwide concern [4]. It is established that high noise levels

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have auditory and non-auditory health impacts on human health. High noise exposure causes auditory issues such as hearing loss, tinnitus and non-auditory issues like sleep deprivation, annoyance and cardiovascular diseases. The prevalence of these negative impacts is highly evident in noise climate with high traffic noise levels mainly contributed through heterogenous transportation systems [5]. In the Indian context, traffic noise is one of the major noise sources, with excessive honking and engine noise, leading to high noise pollution in an urban context.

The perception of acoustic environment can be explained by the model of soundscape perception which is based on the circumplex model of affect [6]. This model is dependent on two principle independent dimensions i.e., 'eventfulness' and 'pleasantness'. With further studies in environmental psychology additional dimensions such as 'appropriateness' [7], 'dynamism', 'monotony' [8] and 'vibrancy' were introduced [9]. The soundscape perception model has been developed to the universal standard in ISO 12913-2,3 which provides a common questioner with different response scale for soundscape evaluation, providing a reliable technique [10, 11]. However it is observed that different socio-demographic factors, cultural associations and noise sensitivity impacts the assessment of soundscape effectively [12, 13].

Studies have explored the association between noise exposure and its impact on health subjectively. A large amount of research efforts has been devoted to decoding the impact of noise by analysing the physiological signals as a response to auditory stimuli, which can change the emotional and physiological responses of listeners. A strong effect of rural and urban soundscapes was analysed on heart rate and human brain waves [14] In another study, it was found that noise annoyance due to traffic noise increases heart rate [15], whereas heart rate decreases when exposed to natural sound [16]. In recent findings impact of noise was found on human brain waves, leading to uncomfortable and annoying state [17, 18]. Change in alpha and beta band was noted due to high noise exposure [16]. While performing assessment of positive soundscapes, Alvarsson et al. noticed that there is faster stress recovery (indicated by SCL) for listeners when exposed to sound from a natural environment [19]. Exposure to pleasant soundscapes have helped listeners to recover from stress [20]. A positive association with eventful soundscape and increase in skin conductance was analysed by Bradley et al. [21]. Another study found that the interactive effect of visual-only VR and nature sounds had a parasympathetic effect on stress recovery [22]. Significant changes in visceral, somatic and central system including changes in heart rate, skin conductance and muscle tone in response to distinctive auditory stimuli is observed [21]. Erfanian et al. have pointed out the need for physiological studies in a laboratory setting using binaural or ambisonic recordings to assess the physiological changes induced due to noise exposure [23]. Physiological changes associated with soundscapes are frequently assessed using electrocardiography (ECG), Galvanic skin response (GSR), Functional magnetic resonance imaging (fMRI) and Electroencephalography (EEG) [19, 24–26]. This study follows an alternative dimension to analyse the effect of soundscapes using physiological signals.

ECG is one of the techniques used for assessing the impact of auditory stimuli on human physiology. One of the initial studies done by Bradly and Lang established that with an increase in loudness level, particularly unpleasant music, there is a decrease in heart rate [21]. Heart rate variability (HRV) is a non-invasive technique for the investigation of the Autonomic Nervous System (ANS) which is divided into two sections, the sympathetic nervous system (SNS) and the parasympathetic nervous system (PNS). HRV is a measure of variation in time between each heartbeat analysed using time and frequency domain parameters. Inter beat interval (IBI), the interval between heartbeats, is measured using time-domain indices of HRV, which quantify the degree of variability. The IBI of normal sinus beats' standard deviation (SDNN) is expressed in milliseconds. Both SNS and PNS activity contribute to SDNN, and they have a strong relationship with the power of the VLF (Very Low Frequency), and LF (Low Frequency) bands. SDNN is considered a global parameter to understand HRV. NN50 is the number of adjacent NN intervals that differ from one another by more than 50 ms, and the percentage of adjacent NN intervals differ from each other by more than 50 ms is called pNN50. This parameter closely relates to PNS activity and correlates with HF power [27]. The distribution of absolute or relative power into frequency bands is estimated through frequency-domain measurements. VLF power is assessed for 0.003-0.04 Hz frequency, mainly linked with all-cause mortality, arrhythmic death, and PTSD. The LF power is in the range of 0.04 to 0.15 Hz, produced by PNS and SNS activity. HF band is in the range of 0.15 to 0.40 Hz reflecting parasympathetic activity. Lower HF power is correlated with stress, panic, anxiety, or worry [27]. LF/HF Ratio represents the relative balance between sympathetic and parasympathetic tones [28].

Multiple studies have explored noise exposure and change in HRV parameters [28–30]. In a recent study, noise is associated with decreased HRV and imbalanced cardiac autonomic function [31]. Research by Biel et al. has established the relation between acute traffic noise exposure

and cardiovascular issues [32]. In another study by Buregeya et al., it is observed that short term noise exposure doesn't have a significant impact on HRV, but one percent of total noise dosage is associated with an increase in variation of HR. Similar studies based on short- term noise exposure have mentioned the change in HRV due to noise exposure causing imbalanced autonomic nervous system activity which can increase the risk of cardiovascular issues [33, 34]. In addition, noise has also been shown to have an impact on neurological, endocrine, cardiovascular, and respiratory systems of the body [35].

Noise sensitivity is one of the major parameters that need to consider while assessing the impact of noise exposure on listeners' physiology and psychology [36]. It is observed that the subjective annoyance level for a person with high sensitivity is more than for a person with low noise sensitivity. Highly noise- sensitive people have been reported to perceive higher stress [37] and sleep disorders [38]. Most importantly, it has been studied that highly noise- sensitive people are at greater risk of cardiovascular issues [39, 40]. To measure noise sensitivity, Weinstein (1978) has developed a standard questionnaire which provides a sensitivity score based on daily reactions towards normal environmental noise exposure [41]. There are few studies which analyses the impact of noise sensitivity level on change in HRV due to noise exposure. An epidemiological study based on long- term noise exposure has reported high HR for people who are highly sensitive to noise than low noise sensitive people [42]. In another study, a significant difference between noise- sensitive groups is noticed for change in HRV [43]. A recent study by Cai et al. has attempted to study the variation in HRV for noise-sensitive groups, but no significant difference has been reported [44].

This paper evaluates the change in HRV for people with different noise sensitivity. ECG signals are used as a physiological response to different soundscapes scenarios presented in a listening experiment. Heart rate variability analysis is adopted to study the changes in various time and frequency domain HRV parameters. The objectives of the study are to (a) analyse the overall change in HRV parameters for different soundscapes exposures and (b) evaluates the change in HRV for high and low noise sensitive groups. This study considers soundscape in heterogenous traffic noise scenarios.

2 Method

2.1 Onsite data recording

Soundscape data was collected in New-Delhi, India. The collection of data was based on the type of land use, street type, traffic intensity and major activities around the recording location. Total 40 locations were selected for the data collection. Data collection process includes the collection of binaural recordings in wave format, sound level meter recordings and video recordings. In this analysis, binaural recordings were used for the analysis. Sound level meter and binaural microphone were mounted on a tripod at a height of 1.5 m, away from any reflecting surface at the measurement location. The data was collected in March 2021, during day time between 9 am to 7 pm. The locations and time period were based on Central Pollution Control Board (CPCB) environmental noise data for multiple locations across the city. Based on this data, noisy and quiet locations were identified.

Audio recordings

Audio recordings were collected in wave file format at the sampling frequency of 48000 Hz, length of each soundscape stimuli used for the experiment was 15 s. These collected binaural audio wave files were processed in MATLAB Audio Toolbox [45] for extracting the psychoacoustic parameters. Four basic psychoacoustic parameters i.e., loudness (N5), Sharpness (S), Fluctuation strength (Fls) and Roughness (R) were extracted for analysis. The psychoacoustic annoyance defined by Zwicker [46] was calculated for each soundscape stimuli. Loudness (N) is a subjective interpretation of sound pressure level or intensity of the sound measure in sones and N5 percentile is the loudness below which is 95% of the reported loudness. The perception of high frequency sound is known as sharpness. It is determined using the waveform of the sound signal, which is the estimated 1/3 octave band spectrum over the frequency range of 25 Hz to 12.5 kHz. Roughness is the fast amplitude modulation of a sound's perceptual effect (15–300 Hz; measured in aspers). Whereas, Fluctuation strength (Fls) is the slower (up to 20 Hz) amplitude modulation of a sound and it is measured in vacils [46].

2.2 Listening experiment

Listening experiment was conducted at the Indian Institute of Technology, Roorkee, in a quiet chamber with RT_{60} 1.1 s. The temperature of the room was constant without any movement of people inside the room. A total of 25 par-

ticipants participated in the experiment, with 9 females and 16 males, between the age group of 20-27 years. All participants were university students pursuing graduate or postgraduate studies. Participants were invited based on inclusion and exclusion criteria. Participants with any hearing or heart- related disorder were not allowed for the experiment. Moreover, persons suffering from any psychological issues such as depression, post- traumatic disorder was not included in the experiment. Every participant has reported normal hearing (subjectively). After reading the participant information sheet and signing the consent form, participants were asked to enter the listening room. This experiment is conducted according to Indian Council of Medical Research, National Ethical guidelines.

Experimental procedure:

At the start of the experiment, participants were asked to fill out the Weinstein Noise Sensitivity Scale survey to assess their noise sensitivity [41]. This experiment involved the collection of three biophysiological signals i.e., Electrocardiogram (ECG), Electroencephalogram (EEG) and Galvanic Skin Response (GSR). EMOTIV epoch+ EEG headset was used to collect EEG data and the SHIMMER Sensing device was used to collect GSR and ECG data. These sensors were used to detect the changes in the biophysiological signals of participants due to exposure of soundscape stimuli. In this paper, changes noticed in ECG signals are reported.

All three sensors were placed on the participant's body (Figure 1). Participants were provided with flat response noise cancellation headphones (Etymotic ER4). LED screen was placed in front of participants to view the experiment procedure, basic instructions and its progress, which was also remotely handled by researcher in the control area.

Before starting the actual experiment, a dummy experiment was conducted with one stimuli, where it was confirmed that all sensors were working with 100% contact quality. The handiness of setup and participant's comfort were also checked in the dummy experiment. Before presenting the first stimuli, the baseline measurement was done for 30 s. This baseline data was used to calculate the relative change in these signals.

The experiment was divided into 4 sets, in each set, 10 stimuli were presented, with a length of 15 s. The intra-stimulus gap was maintained at random between 15-30 s. The presentation of all 40 stimuli was also randomised. Between each set, participants were provided with a short break, during which they were allowed to take rest but were not allowed to leave the listening room. While presenting the stimuli, participants were asked to focus on the white LED screen with a green dot in the centre. They were instructed to make no to minimal movement during the experiment period, specially while listening to the stimuli. After listening to all stimuli, all sensors were removed from the participants' body and they were allowed to take rest for 10-15 min. During this period, they were also allowed to leave the listening room.

2.3 ECG response

The Shimmer ECG sensor was used to ECG collect data in the experiment [47, 48]. The data was collected at the sampling rate of 128 Hz. In this experiment, four bipolar limb leads were used, which were pasted at the left arm (LA), right arm (RA), left leg (LL) and right leg (RL). Shimmer ConsensysPRO Software was used to acquire the ECG data and perform the basic processing.

ECG data processing

After collecting the session long data for all 25 participants, raw data is exported to Kubios software for preprocessing and further analysis. The scientific validity of this software for performing ECG data is also confirmed by multiple studies [49,50]. For this analysis, data from LA-RA electrode is used. A noise detection algorithm is used to remove the unwanted noise signals in the dataset. Further, an automatic beat correction algorithm [51] is used with an acceptance threshold of 5% for correcting the missed, extra and misaligned beats from RR interval data. After conducting the pre-processing of the signal, the data is analysed automatically for each stimulus interval to provide different time and frequency domain parameters. The parameters used in this study are summarised in Table 1.

Dataset preparation

ECG responses of 25 participants were collected and processed using kubios software. The data of one participant was removed due to a few anomalies. Finally, data of 24 participants was used for the final analysis. The relative change with respect to the baseline measurement was calculated using Eq. (1)

$$Relative\ change\ (\%) = \frac{Raw\ data - Baseline\ Data}{Baseline\ Data} \times 100 \quad (1)$$

The HRV parameter for each stimulus corresponding to each participant was extracted, which was further averaged across all 24 participants to get the overall mean of each HRV parameter for each soundscape stimuli.

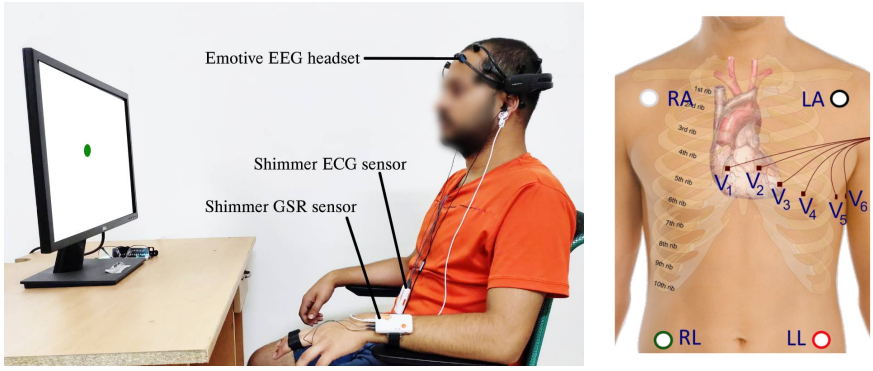


Figure 1: Experimental setup in a listening room (left), Placement of ECG electrodes on participant body (right) (image adopted with permission from shimmersensing.com)

Table 1: HRV parameters considered for analysis.

Time domain HRV parameters	
Min HR (1/min)	Minimum Heart Rate
SDNN (ms)	Standard deviation of RR intervals
NN_{50} (beats)	Number of successive RR interval pairs that differ more than 50 ms
pNN_{50}(%)	NN50 divided by the total number of RR intervals
Frequency domain HRV parameters	
VLF (ms2)	Absolute VLF power (FFT), (0.00-0.04 Hz)
LF (ms2)	Absolute LF power (FFT), (0.04-0.15 Hz)
HF (ms2)	Absolute HF power (FFT), (0.15-0.4 Hz)
LF/HF	Ratio between LF and HF band powers

3 Results

3.1 Characterization of soundscape stimuli

This section presents the variation in psychoacoustic indicators of each soundscape stimuli used in listening experiment. Table 2 presents the statistical summary of psychoacoustic indicators analysed for 40 locations.

LAeq for these locations falls between 57.36 dB (P40) to 85.09 dB (P34). The average noise level i.e. LAeq 78.90 dB is observed, with first and third quartile range of 74.21 dB and 83.10 dB respectively. This indicates that there is a combination of noisy and silent locations in the study, where majority of the signal are relatively loud. N5 is considered as an indicator to define the loudness of the stimuli. Considering the overall time of recording, the maximum loudness level of 53.86 sones is observed at location (P40), and the minimum level of 5.28 recorded at location (P34). Location P40 has a maximum overall loudness level of 52.44 sones and location P34 has the lowest overall loudness of 7.66 sones calculated for 15 s. The average level of loudness across all location is 34.19 sones, and half of the stimuli lies between the 24.66 sones (1st quartile) to 45.66 sones (3rd quartile). The sharpness represents high- frequency content of the signal. The sharpness level for 40 stimuli lies between 1.22 (P24) to 2.27 (P03) acum. Maximum overall sharpness is observed at (P24) 1.31 acum, and the lowest overall sharpness level at location (P34) with 1.89 acum. Across all locations mean sharpness of 1.64 acum is observed, with 1.59 acum (1st quartile) and 1.70 acum (3rd quartile). Fluctuation strength (Fls), which is a factor of amplitude modulation up to 20Hz, was in the range of 0.010 (P23) to 0.060 (P26) vacils with an average of 0.033 vacils. Also, the range of roughness was between 0.020 (P15) to 0.130 (P31) aspers, with a mean value of 0.091 aspers. The psychoacoustic annoyance, which is calculated using the above four parameters, has the highest value of 53.75 at (P40) and the lowest value of 7.94 at (P34). This shows that among all locations, location P34 is the least annoyed and P40 is the highly annoyed location.

Figure 2 presents the variation of psychoacoustic parameters at each location, using a boxplot. It is observed that locations P1-P2, P15, P33 and P34 are among a few locations with relatively low levels of loudness and standard deviation within the range of 1.00 to 2.11 sones.

Table 2: Statistical summary of psychoacoustics indicators.

	LAeq (dB)	N5 (sones)	Sharpness (acum)	Fluctuation Strength (vacil)	Roughness (asper)	Psychoacoustic Annoyance
Min.	57.36	7.66	1.310	0.010	0.020	7.94
1st Qu.	74.21	24.66	1.597	0.020	0.072	25.73
Median	79.36	35.23	1.635	0.030	0.090	36.70
Mean	78.93	34.19	1.647	0.033	0.091	35.44
3rd Qu.	83.10	45.66	1.710	0.042	0.110	47.34
Max.	85.09	52.44	1.890	0.060	0.130	53.75

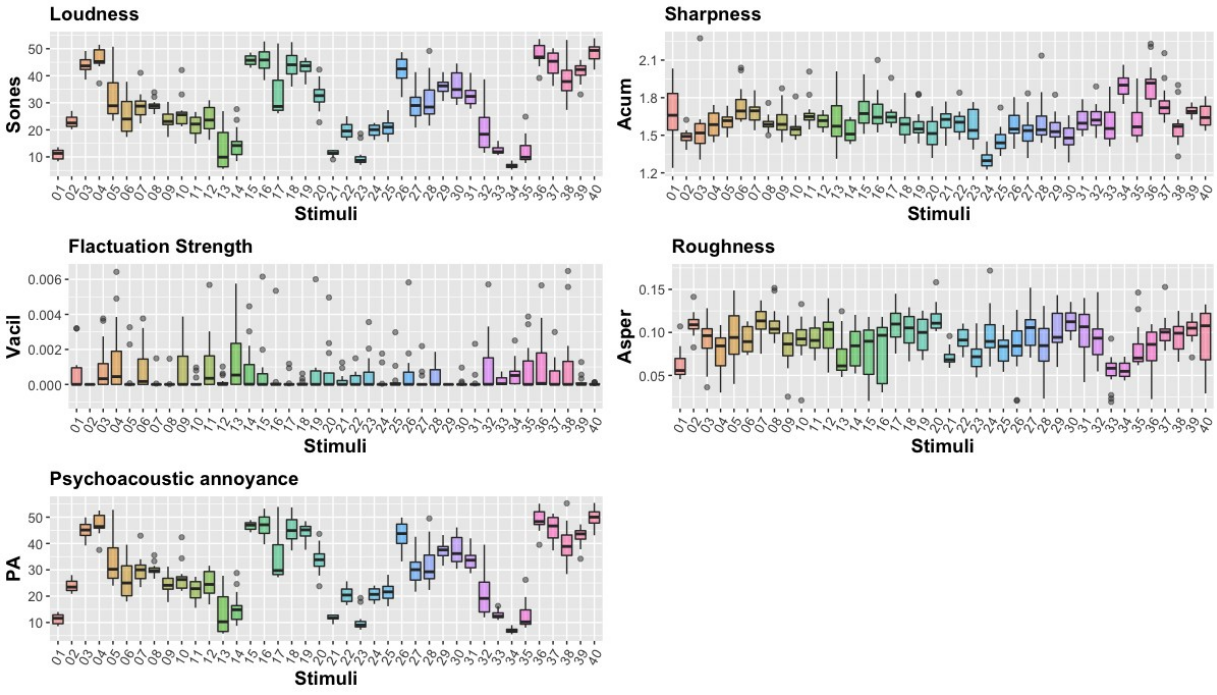


Figure 2: Variation of psychoacoustic indicators across all 40 stimuli.

These soundscape stimuli can be classified as silent and monotonous signals without much fluctuation in signal energy. They are mainly recorded in quiet areas with limited traffic and no honking. The natural sound and vehicles' engine noise were major dominant noise signals at very low amplitude. The maximum variation of loudness was observed at locations P5, P6, P13, P17, P28, P32 and P38, with the standard deviation between 6.92 to 8.76 sones. These locations were recorded predominantly in residential areas. These wide variations in the loudness level were observed due to intermittent honking, bike thumping noise, or people talking loudly during the recording time. The average loudness level of these signals was not as high as the level found in a few locations with a continuous loud noise. The locations P3, P4, P15, P16, P18, P19 and P36-40 have an average loudness level of 44.55 sones with a standard deviation of 3.93 sones. It shows that these locations have a continuous loudness level with a less temporal variation. Such stimuli were mainly recorded in the area with heavy traffic, continuous honking and market areas.

Considering the variation in the sharpness level of selected locations, the average sharpness across location is 1.61 acum. Stimuli P1, P3, P13 and P28 have a high standard deviation between 0.18 to 0.23 acum. Stimuli P2, P5, P8, P12, P24 and P39 have very low spread of change in sharpness during the recording period, with a standard deviation in the range of 0.03 to 0.06 acum. These soundscape scenarios are characterised by a continuous movement of less traffic and are predominantly dominated by engine noise. Stimuli P34 and P36 have an average sharpness value of 1.89 and 1.90 acum, respectively. Also, it is

noted that the standard deviation of these stimuli is 0.09 and 0.15 acum respectively.

The fluctuation strength and Roughness of the signal were analysed for the variation across mean values. Stimuli P25 and P33 have the highest mean of 0.0311 vacil and a standard deviation of 0.082 vacil. Rest of the stimuli have an average fluctuation strength of 0.0036 vacil and a standard deviation of 0.0086 vacil. Also, highest level of roughness was observed in stimuli P2, P7, P8, P17, P20 and P30 with an average of 0.111 asper and standard deviation of 0.016 asper. Variation in psychoacoustic annoyance is calculated for all stimuli to assess the quality of sound. Stimuli P3, P4, P15, P16, P18, P36, P37 and P40 have PA level above 45. The lowest level of PA was observed at P23 and P34, with PA of 10.23 and 13.07, respectively. A high level of PA was observed in the area of traffic congestion, with a high level of honking and an extensive level of engine acceleration.

3.2 Clustering of soundscapes

Considering the complexity of soundscape scenarios and its variation in amplitudes of psychoacoustic indicators, clustering of these stimuli was performed. The psychoacoustic parameters N5, sharpness, roughness, fluctuation strength and psychoacoustic annoyance were used for the clustering. The dataset was scaled and the gap statistic algorithm [52] was used to find the optimum number of clusters, which was set to three. K-means algorithm is used to form the clusters using R and package factoextra [53]. Figure 3 presents the clustered data based on the two primary dimensions for the formed clusters. There was a major contribution of N5 and Annoyance level in dimension 1 and Sharpness and Roughness in Dimension 2. As Annoyance is also characterized as a function of loudness perception and sound quality [46] Dimension 1 can be termed as a function of sound quality and Dimension 2 is considered as a function of spectro-temporal variation. Also, while comparing the dimension with soundscape model of perception [2, 9], Dimension 1 is along the diagonal axis of clam-chaotic and Dimension 2 is along vibrant-monotonous diagonal axis. Most of the locations falls under calm, chaotic and vibrant soundscape categories, no soundscapes scenarios were into monotonous soundscapes.

Cluster-1 consists of stimuli total of 16 stimuli which belong to loud soundscape scenarios. Traffic noise, honking, mechanical and human noise are major sources in this cluster, having highest noise levels. These soundscape scenarios were recorded in market place areas and sub-

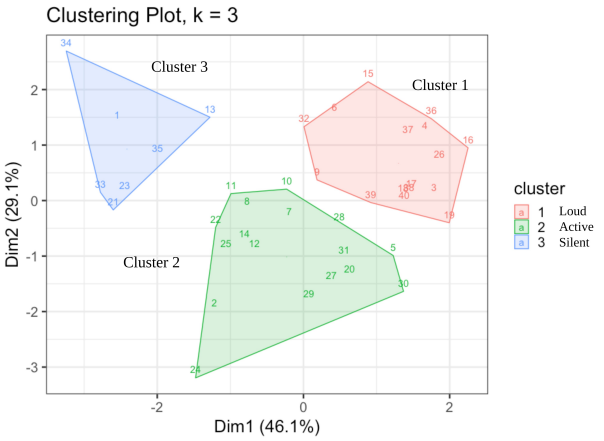


Figure 3: Clustering of noise stimuli in two primary dimensions using k-means clustering method.

arterial streets with higher traffic congestion. This cluster is mainly a combination of highly eventful and unpleasant scenarios leading to a chaotic environment. Cluster-2 has 17 stimuli with a relatively less loud soundscape stimuli. The intermittent honking, constant traffic flow and natural sound are few dominant sources, increasing the vibrancy of the space. These soundscape are highly eventful with a combination of pleasant and sometimes unpleasant scenarios. Roughness level for this cluster is maximum which indicates high vibrant scenario [2, 9]. Cluster-3 consist of 7 stimuli with minimal to no traffic movement. This cluster was mainly silent with a few sound events from trees and birds. These soundscapes can be considered as calm or relaxing and falls between less eventful and pleasant dimensions. Based on the loudness level, sound quality and associated soundscape descriptors these three clusters are termed as 'Loud', 'Active' and 'Silent' cluster respectively. The loud cluster represents chaotic scenarios, the active cluster shows vibrant scenarios, and the Silent shows calm soundscape scenarios. Representative images for each cluster are presented in Figure 4.



Figure 4: Representative images for three clusters.

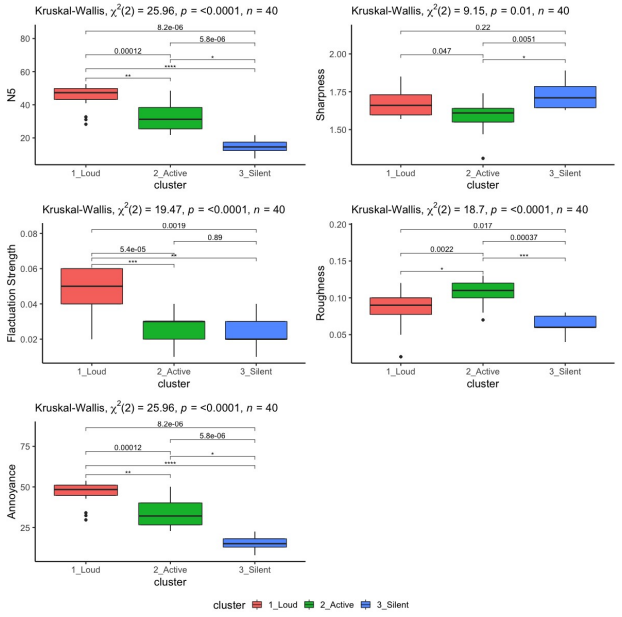


Figure 5: Variation in psychoacoustic parameters for three clusters.

Figure 5 presents the variation of indicators in different clusters. Kruskal-Wallis test is conducted to find out the variation among clusters across all indicators. It is observed that across all indicators, the variation among clusters is statistically significant. For each indicator, the difference between cluster was significant with p<0.05.

Table 3 presents the statistical summary of the clusters formed. The loud cluster has an average N_5 of 44.46 sones (σ=7.49 sones), whereas active and silent clusters have mean loudness of 32.53 and 14.78 sones and Standard deviation of 1.59 and 1.73 sones, respectively. The mean value of sharpness for the loud cluster is 1.67 acum (σ=0.08 acum), for active cluster 1.59 acum (σ=0.10 acum) and for silent cluster 1.73 acum (σ=0.10 acum). The sharpness level of the silent cluster is high compared to the loud cluster with a higher standard deviation (8.04 acum). Mean value for the fluctuation strength of the loud and active cluster is 0.048 and 0.025 vacil respectively. Roughness level for the loud cluster was 0.085 asper (σ=0.024 asper) and for silent cluster it is reported at 0.064 asper (σ=0.014 asper). Active cluster have highest mean roughness of 0.109 asper (σ=0.017 asper). Psychoacoustic annoyance level corresponds with the loudness of stimuli. Loud cluster has mean PA of 45.92 (σ=7.51), the active cluster has a PA of 33.86 (σ=8.32) and the silent cluster have reported PA of 15.33 (σ=4.96).

In auditory studies, Just Noticeable Difference (JND), is the minimum amount by which the intensity of a stimulus must change for a change in sensory experience to be noticeable. The just noticeable difference in loudness increases with increases with an increase in the SPL of the signal. The JND for sharpness have a small variation with an increase in the overall sharpness of the signal. The change in loudness is perceived by participants with an increase in loudness by 0.8 sones. Whereas the difference of 0.04 acum is signal sharpness is noticeable by the listeners [54]. The mean difference of all psychoacoustic parameters is above JND values for all clusters. Hence, it can be said that the changes in HRV parameters are caused by changes in all psychoacoustic parameters that were looked at.

3.3 Change in HRV for soundscapes clusters

This section analyses the change in HRV parameters due to exposure to different soundscape stimuli. Two types of parameters, time domain and frequency domain, are analysed for three developed clusters. Figure 6 presents the variation of time domain HRV parameters for Loud and active, and silent soundscapes stimuli. The mean relative change (%) with respect to the baseline, averaged across all participants, is used for the analysis.

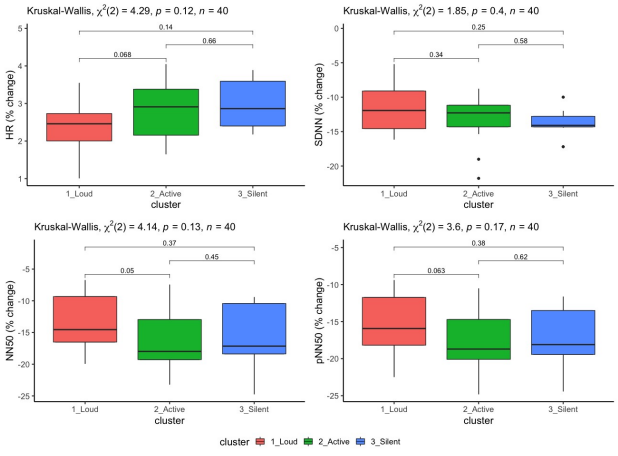


Figure 6: Relative change (%) in time domain HRV parameters for different soundscape categories.

Relative change in min HR significantly decreases with an increase in noise levels. Loud cluster has highest reduction in HR level. Kruskal-Wallies test suggest no significant difference across clusters (p=0.12). SDNN show reduction of 11.46% in loud cluster, 13.26% for active cluster and 13.64% for silent cluster is noted. No significant variation between the cluster is noted (p=0.4).

Change in NN_{50} is checked for the given groups. NN_{50} increases by 13.42% for loud cluster, for the active cluster

Table 3: Statistical summary of formed clusters

Cluster No.	No of Stimuli	Type	N (sones)		S(acum)		Fls (vacil)		R (asper)		PA	
			Mean	STD	Mean	STD	Mean	STD	Mean	STD	Mean	STD
1	16	Loud	44.46	7.49	1.67	0.08	0.048	0.013	0.085	0.024	45.92	7.51
2	17	Active	32.53	8.04	1.59	0.10	0.025	0.009	0.109	0.017	33.86	8.32
3	07	Silent	14.78	4.81	1.73	0.10	0.024	0.010	0.064	0.014	15.33	4.96

it increases by 16.46% and for the silent cluster increase of 15.56% is noted. The variation among the three clusters is analysed and no significant variation was found among the clusters (p=0.13), whereas, the Wilcoxon test conducted between active and loud clusters shows statistically significant variation with p=0.05.

Change in pNN50 is checked for the scenarios, for the loud cluster, pNN50 is reduced by 15.28%, for the active cluster, the reduction happens by 18.21% and for the silent cluster, the pNN50 is reduced by 17.14%. No significant variation among the clusters is observed through Kruskal Wallis test (p=0.17). The Wilcoxon test between loud and active cluster have a marginally significant variation with p=0.06.

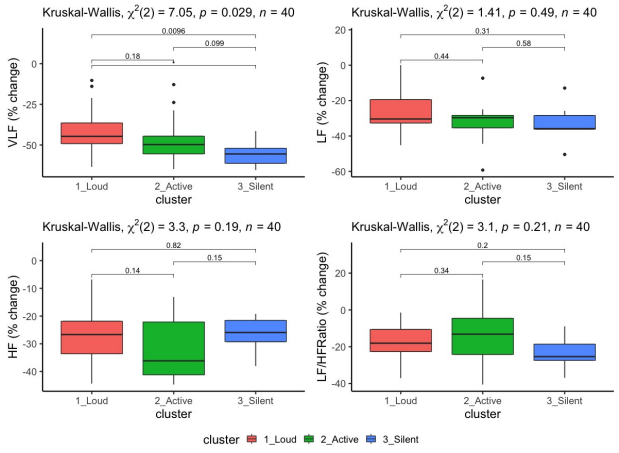


Figure 7: Relative Change (%) in frequency domain HRV parameters for different soundscape categories.

Figure 7 presents the variation of HRV parameters in the frequency domain. The rate of change of very high frequency (VLF) power for the loud cluster is -40.68%, and more reduction of VLF is noticed for the active cluster -46.57%. Following the trend in reduction, the silent cluster has a relative change of -55.55%. The krusakal-wallis test suggested significant variation among the clusters (p=0.02). Wilcoxon test conducted between pairs of clusters shows a statistically significant variation between loud and silent cluster (p<0.01). No significant relation is noticed for other pairs of clusters. The change in low- frequency power of the loud cluster is -26.94%, followed by -32.07% reduction in the active cluster and -32.62% reduction in the silent cluster. No significant variation is noticed among the clusters (p=0.33) and pairs of clusters for LF (p>0.05). The relative change in power of high frequency for the loud cluster is -26.87%, for the active cluster -32.36% and for the silent cluster -26.39% is noted, with no significant difference. The power of LF and HF ratio is reduced by 17.48% for the loud cluster, 11.87 for the active cluster and 23.30 for the silent cluster. The tests conducted for analysing the significant variation among clusters have reported negative results.

3.4 Change in HRV for noise sensitive groups

In this section changes in HRV parameters for different clusters are assessed with respect to the noise sensitivity level of the participants. The sensitivity score of the participant is calculated through Weinstein Noise Sensitivity Scale. The Z-score is calculated for the final noise sensitivity level. Participants having a sensitivity level (SL) above the mean are deemed as 'Highly Sensitive' and participants below the mean level are considered as 'Low Sensitive'.

Figure 8 presents the relative change in time domain HRV parameters when considering SL groups for loud, active and silent soundscape stimuli. The change in HR was found to have a significant difference between high and low sensitivity groups for all three clusters (Wilcoxon test, p<0.05). Reduction in HR was noticed for people with low noise sensitivity than high noise sensitivity.

Change in SDNN is analysed for the loud cluster, for the high SL group, the reduction in SDNN is -13.11% and for the low SL group, it is -7.12%, showing a statistically significant difference (p<0.05). Change in SDNN for the active cluster for the high SL group is -14.04% and for the low SL group is -8.69%, which is significantly different, p<0.05. The change in SDNN high and low SL groups in

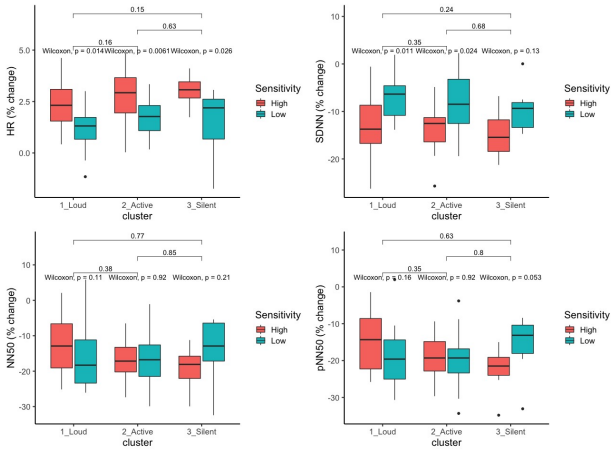


Figure 8: Relative change (%) in time domain HRV parameters for high and low noise sensitivity group.

the silent cluster have a % change of -14.84% and -9.58%, respectively. This variation is not significant, p>0.05. From the result, it is noted that sensitivity levels have an impact on loud and active clusters.

The change in NN_{50} for the high and low SL group in the loud cluster is -12.67 % and -16.87%, respectively. For the active cluster, a reduction of NN_{50} by -12.67% and -16.87% is noticed for high and low SL groups. Whereas, the change of -19.27% and -13.99% is noted for high and low SL groups, respectively in the Silent cluster. No significant variation is noticed for change in NN_{50} due to the changes in the sensitivity level of the participants. The change in pNN50 is checked for loud, active and silent clusters for high and low SL groups. No significant variation is noted in the change of pNN50 due to sensitivity level, across all groups.

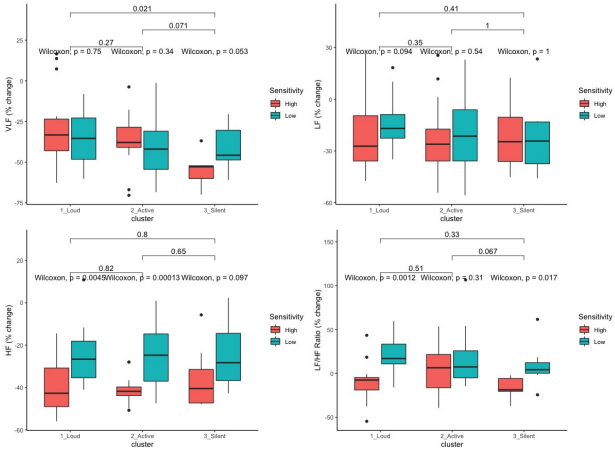


Figure 9: Relative change (%) in frequency domain HRV parameters for high and low noise sensitivity group.

Figure 9 presents the change in HRV parameters in the frequency domain for high and low sensitivity groups. For VLF power, there is no significant variation in the VLF power of the high SL (-29.58%) and low SL group (-35.74%) in the loud cluster. Also, for active and silent cluster no significant difference in change of VLF power is noticed for sensitivity groups. While comparing the clusters, it is noted that loud and silent cluster have a significant variation, p=0.02. Variation in LF power does not differ significantly due to changes in sensitivity level, for all three clusters. The reduction of LF power in the high SL group for all clusters is marginally higher, but not statistically significant to consider the impact. Modulation in HF power for loud cluster in the high SL group is -39.93% and for the low SL group is -25.30% which is statistically different (p<0.05). Statistically significant change in low and high SL groups for active cluster is -41.29% and -25.05%, respectively (p<0.05). There is no significant variation among SL groups for the silent cluster (p=0.09). The difference in LF/HF power for the loud cluster in high and low SL groups is significantly varying, with -9.91% and 19.5%, respectively (p<0.05). The variation in SL groups for the active cluster is not significant in nature. For Silent clusters, the variation is significant for high SL (-15.83%) and low SL (9.46%) with p<0.05.

4 Discussion

4.1 Soundscapes and its perception

The soundscape stimuli considered in the listening experiment have varying characters. Although these stimuli have been clustered into three categories, it is still important to discuss the noise climate of New Delhi for better insight. High intensity of vehicle honking, engine noise and human noise are primary dominant sources. This is due to the heterogeneous nature of traffic with different types of vehicles, speed, acceleration, combination of motorised and non-motorised vehicles and non-adherence to lane system [55]. This scenario results in congestion on streets leading to excessive honking as the people try to get around. The locations with continuous traffic congestion, hawkers talking on the street, and continuous high noise levels, which are unbearable and classified as a loud cluster in this study. This cluster consists of traffic congestion scenarios on street junctions, market areas adjacent to local streets etc. A group of locations with relatively low congestion and moderate noise levels but a high level of intermittent honking, which is sometimes unexpected, is

termed as an active cluster, it can be also considered as vibrant scenarios. Silent cluster consists of the locations with low noise levels, free- flowing traffic and very less honking. These locations are mainly found in residential areas, parks and free- flowing arterial streets with less traffic.

Table 3 presents the cluster attributes from which it is perceived that N5 for loud cluster active and silent cluster is 44.46, 32.53 and 14.78 sones, respectively. This high level of loudness is mainly associated with traffic noise in an urban setting and leads to a high level of annoyance [56]. It is seen that the sharpness level, which is the sensation of high frequency content in noise stimuli, is very high in active cluster as compared to the loud cluster [57]. Hong et al. make a similar observation, noting that in moderately noisy soundscape circumstances, the sharpness level (bird noise) is noticed more prominently. In this study, the possible explanation is that the occasional honking, which contains a high amount of energy at high- frequency band leads to high sharpness, which is heard more clearly in environments with less ambient noise. Honking is considered as an unpleasant and annoying part of the soundscape [58], leading to a negative impression. The high fluctuation strength of loud cluster is contributed by human voice, music and footsteps and vehicle noise in urban areas. Whereas the high roughness level is contributed by mechanical sound [59] and also human sound [60]. The perception of loud cluster, while considering their psychoacoustic parameters, can be termed as annoying and unpleasant. As examined by Brink et al. that the intermittent noise from road traffic has a greater impact on the annoyance level of the listener [61]. The presence of an intermittent honking pattern in the active cluster sometimes cause a high or equal level of irritation and annoyance when compared with a loud cluster.

4.2 Soundscapes, HRV and noise sensitivity

Change in HRV parameters in time and frequency domain is analysed for loud, active and silent soundscape stimuli. Time domain parameters mean HR, SDNN, NN50 and PNN50 are analysed for the change due to exposure to sound stimuli. It is noticed that heart rate increases with an increase in loudness. This may be due to exposure to loud abrupt honking noise, which leads to acoustic startle reflex, and the effect induced by abrupt high noise levels on cardiovascular functioning [62]. While analysing the HR for the high and low SL group, it is noted that, the heart rate is higher for the high SL group. SDNN is the global parameter for HRV and representative of the autonomic function of the heart. It represents sympathetic and parasympathetic activity. No significant variation in SDNN is noticed across clusters. SDNN has a significant correlation ($r=0.36$, $p<0.05$) with the loudness level of stimuli, indicating that with an increase in loudness, SDNN increases. While assessing the change in SDNN for sensitivity groups, it is found that SDNN decreases for the high SL group as compared with the low SL group. The variation between SDNN for high and low SL groups in the active and loud clusters is significant. There is no significant change between groups for the silent cluster. This indicated that lower noise levels do not affect the global HRV, but it significantly affects higher noise levels among highly noise-sensitive people. This decrease in SDNN is also found by Amaral et al., where exposure to heavy metal music (80-90 dB) was presented to the listeners [63]. The reduction in HRV indicates the poor performance of cardiovascular functions, and an increase in HRV indicates improvement in cardiovascular function [63]. This suggests that people with high noise sensitivity have a negative impact on cardiovascular function due to noise exposure.

pNN50 is another parameter which highly correlates with PNS activity [27]. It is noticed that pNN50 for the loud cluster is higher compared to the active and silent cluster. There is no significant variation observed among the cluster, but a marginal variation is observed between active and loud cluster. The sensitivity level has no impact on pNN50 for all types of soundscape stimuli. Whereas, a significant difference between the percent change of NN50 for the loud and active clusters. It shows that NN50 reduces considerably for the active cluster when compared with the loud cluster. A similar trend is noticed for NN50 in terms of the sensitive group, where no difference is noted between the high and low sensitivity groups. pNN50 is related to parasympathetic nerves in the heart, this indicates major parasympathetic activity while listening to loud stimuli.

In the frequency domain HRV parameters, VLF power has a significant difference between loud and silent clusters. VLF power is reduced to the maximum for the silent cluster, followed by the active and loud cluster. Also, a significant positive correlation is observed between N5 and VLF power ($r=0.42$, $p<0.05$). LF, HF and LH/HF ratios have not shown any significant difference for all three clusters. There is an increase in LF and LF/LH ratio with an increase in loudness level, similar results are noted by Sim et al. [64]. LF power is an indicator for sympathetic and parasympathetic nervous system activity [65], indicating that while listening to different soundscape stimuli participants have experienced a change in comfort, which leads to annoyance. HF power is mediated by the parasympa-

thetic nervous system [66], and changes in the LF:HF ratio provides information about the balance between sympathetic and parasympathetic modulations [67]. In this study, the LF power is relatively lower for silent categories when compared with loud and active clusters, suggesting that the listener has experienced less discomfort due to silent soundscape stimuli. While comparing the sensitivity group, for change in VLF and LF no significant difference between high and low SL group is noted, indicating no considerable change in sympathetic nervous system activity. Participants with high SL level have showed a significant decrease in HF power for the loud and active cluster, which show change in parasympathetic activity. LF/HF ratio reduced significantly for the loud and silent clusters for the high SL group. A similar positive association with LF/HF ratio with noise levels is found by [34, 67, 68].

Overall there is no discernible change in HRV for the three studied clusters, except min HR. This may be due to, habituation to high noise levels for listeners with low noise sensitivity. When these changes are analysed with respect to the sensitivity level of participants, many statistically significant changes are observed between the high SL and low SL groups. It is observed that for participants with high sensitivity, the SDNN decreases, and the heart rate increases considerably. The change in HF and LF/HF is also noted to have significant variation for highly noise sensitive people. This shows that noise sensitivity is an important parameter to assess while studying the change in physiological signals due to noise exposure. Several studies have reported a similar variation based on sensitivity [39, 40]. Our results are in line with the previous studies where a negative change in HRV is noticed [32, 34, 69, 70].

A change in HR level and a decrease in SDNN for the high sensitive group shows a decline in HRV, indicating an increased risk of noise exposure for the high noise sensitive group. Such increased risk leads to cardiovascular issues such as hypertension, increased blood pressure and myocardial infarction [71, 72]. Multiple studies have reported a reduction in SDNN with cardiovascular mortality [73–75]. A study conducted by Tsuji et al. [76] reveals that a fall in one standard deviation is associated with a 50% increase in the risk of cardiovascular events over a period of 3.5 years, making the SDNN the strongest predictor of new cardiac events. The imbalanced cardiac autonomic function is observed by decreased HF and pNN50 for the high SL group. In a study conducted by Stansfeld et al., the impact on the autonomic nervous system leads to affect the cardiovascular system [77]. Reduced HRV and sympathetic-parasympathetic imbalance have significant therapeutic implications because they may cause electrical instability and cardiac arrhythmia as the ANS controls the electrical regulation of the heart [28].

Based on previously explained implications about the decreased rate of HRV, it can be assumed that due to exposure to noise, there is a release of stress hormones such as catecholamines and glucocorticoids, and these hormones may both, directly and indirectly, cause problems with the autonomic nervous system [32, 78]. Reduced HRV and unbalanced cardiac autonomic function would result from increased sympathetic neurotransmissions caused by increased stress hormone release [67]. Additional possibilities included psychological responses brought on by annoyance and noise sensitivity levels [78].

Finally, it is summarised that there is an activation of the autonomic nervous system due to exposure to loud soundscape scenarios. This activation can lead to an increased risk of morbidity and mortality. It is also found that noise sensitivity level plays an important role while analysing the HRV changes due to noise exposure. High noise sensitive people are at great risk of cardiovascular issues. It is worthwhile to note that subjects used in this study were not proportionately divided based on gender. Hence, we suggest performing a similar experiment to identify changes in HRV due to noise exposure based on gender, with a proportionate gender ratio. This study is conducted for acute noise exposure, to get a better insight into the long- term impact, replication of this study with a large number of subjects from different occupational backgrounds should be examined for multiple days.

It is necessary to consider current findings in soundscape studies while considering the relevance of noise exposure on human health. People with high noise sensitivity have a significant impact on loud and active clusters was observed. The loud cluster has a combination of high loudness, and the active cluster has high roughness and fluctuation strength. It is required to study the relationship between noise indicators affecting human physiology and the listeners' health. There is a need to design noise standards by considering the impact on the population with high noise sensitivity. The positive impact of soundscapes cannot be ignored; hence it is necessary to transform the soundscapes perception towards a more pleasant and vibrant environment. For instance, introducing natural sound elements in the existing soundscape can reduce stress and other physiological issues. Soundscapes maps can be prepared to identify the hotspots and predict the risk of cardiovascular issues among people living in the vicinity through an interdisciplinary approach. These predictions can help to develop new policies for noise mitigation and soundscape enhancement.

5 Conclusion

In this study, HRV analysis is conducted to analyse changes due to exposure of different soundscape scenarios. For this three clusters i.e. 'loud', 'active' and 'silent' were formed for analysis. There is a significant variation between the psychoacoustic indicators of these clusters. It is observed that min HR and noise levels have a significant relation. The overall variation in SDNN which is considered a most important parameter in HRV have no significant difference among clusters, but a significant decrease in SDNN is observed for high SL groups for loud and active cluster. This decrease in SDNN suggest decrease HRV with imbalanced autonomic nervous system. In frequency domain parameter, no significant changes are observed for overall LF, HF and LF/HF ratio. A significant decrement is observed in HF power and LF/HF ratio for loud and active cluster for high SL group in HF power.

From these results it is concluded that loud noise exposure have an impact on HRV. The unbalanced rate of HRV leads to release of stress hormones which affects ANS. Loud and active traffic noise scenarios are majorly affecting heart rate, SDNN and other frequency domain parameters. The honking scenarios and high traffic congestion are major contributing events for this phenomenon. Exposure to loud noise have a greater impact on people with high noise sensitivity, with an increased risk of cardiovascular issues. There is a need for extensive research focused on long term noise exposure in different occupational settings. It is also required to analyse the impact of noise in gender specific studies. It is essential to analyse the influence of noise pollution on cardiovascular disorders. These findings can help to form robust noise control regulation while considering the impact on vulnerable groups.

Acknowledgement: The author thanks all the participants for participating in the experiment.
Funding information: The research work is supported under Prime Minister Research Fellowship, Ministry of Education, India (PMRF ID: 2800113).
Author contributions: All authors have accepted responsibility for the entire content of this manuscript and approved its submission.
Conflict of interest: The authors state no conflict of interest.

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